# SHA-256 Logic Gates and Silicon Carbide Phonon-Polariton Gate Notes

*Educational engineering reference – not a fabrication protocol or demonstration of a silicon carbide SHA-256 processor.*

## 1. Required Logic Gates and Operations

SHA-256 requires Boolean operations, fixed rotations/shifts, and 32-bit modular addition with correct carry propagation. Fixed rotations and shifts can often be implemented as bit routing rather than arithmetic gates.

**AND:** A AND B

**XOR:** A XOR B; output 1 when inputs differ

**NOT:** NOT A; inverts a bit

**OR:** A OR B

**NAND:** NOT (A AND B); universal Boolean gate

**NOT truth table:**

| A | NOT A |
|---|---|
| 0 | 1 |
| 1 | 0 |

**32-bit modular addition:**

```
(A + B) mod 2^32
Overflow beyond bit 31 is discarded, but carry propagates.
```

$\text{ROTR}^n(x)$: rotate right by n bits, wrapping bits.
x >> n: logical shift right; fill left positions with zeros.

## 2. SHA-256 Boolean Functions

**Choice:**

```
Ch(x, y, z) = (x AND y) XOR ((NOT x) AND z)
```

**Majority:**

```
Maj(x, y, z) =
  (x AND y) XOR (x AND z) XOR (y AND z)
```

## 3. SHA-256 Sigma Functions

```
Sigma0(x) = ROTR^2(x)  XOR ROTR^13(x) XOR ROTR^22(x)
Sigma1(x) = ROTR^6(x)  XOR ROTR^11(x) XOR ROTR^25(x)
sigma0(x) = ROTR^7(x)  XOR ROTR^18(x) XOR (x >> 3)
sigma1(x) = ROTR^17(x) XOR ROTR^19(x) XOR (x >> 10)
```

## 4. How SHA-256 Uses These Operations

SHA-256 processes 512-bit blocks and maintains eight 32-bit state words. Each compression operation uses Choice and Majority functions, sigma functions, 32-bit modular additions, fixed

constants, a message schedule, and 64 rounds per block. A complete circuit also needs reliable state storage, bit routing, timing, signal restoration/fan-out, and carry propagation.

# 5. Building Toward a Physical Silicon Carbide Phonon-Polariton Gate

Development should proceed from a defined physical encoding through NOT/AND/NAND, XOR, a full adder, word-level operations, and finally SHA-256 verification against trusted software references and standard known-answer vectors.

```
STAGE 1: Define polariton encoding, inputs, outputs, threshold, errors.
STAGE 2: Test polariton-input NOT and AND/NAND for every input state.
         NOT: 0 -> 1 and 1 -> 0.
STAGE 3: Demonstrate polariton-input XOR:
         00->0, 01->1, 10->1, 11->0.
STAGE 4: Demonstrate polariton-input full adder:
         SUM  = A XOR B XOR Cin
         COUT = (A AND B) OR (Cin AND (A XOR B))
STAGE 5: Build short polariton carry chains, then 32-bit addition.
STAGE 6: Verify SHA-256 against software reference and
         known-answer vectors.
```

## 5.1 Physical Length and Phonon-Polariton AND-Gate Investigation

A physical length or resonator geometry must control a measurable polariton-to-polariton response that distinguishes all four input combinations; geometry alone does not guarantee nonlinear logic. Candidate mechanisms include polariton propagation delay, geometry-dependent resonance selection in patterned SiC resonators, and an interaction region whose length affects mode overlap and nonlinear (second-order or anharmonic) coupling.

**AND truth table:**

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

For a propagating polariton mode:

```
delta_phi = (2 * pi * f * delta_L) / v
```

where $\Delta\varphi$ is relative polariton phase, $f$ is the polariton frequency, $\Delta L$ is the path-length difference, and $v$ is the polariton phase velocity. For surface phonon-polaritons, $v$ is strongly frequency-dependent and is much lower than the speed of light in vacuum, so this relation cannot be applied to a SiC device without a validated mode, dispersion, and loss model for the specific polytype, geometry, and surrounding dielectric.

A candidate gate must keep the output polariton state low for 00, 01, and 10 while producing a distinguishable high output state for 11. Linear polariton interference may depend on both inputs but does not automatically realize AND logic.

# 6. Physical and Laser Considerations

In common SiC polytypes (for example 4H and 6H), the transverse optical (TO) phonon lies near 796 $cm^{-1}$ (about 24 THz) and the longitudinal optical (LO) phonon near 972 $cm^{-1}$ (about 29 THz); values vary somewhat with polytype and source. Between these frequencies lies the Reststrahlen band (roughly 10.3–12.6 µm in wavelength), where the real part of the permittivity is negative and surface phonon-polaritons can be supported. The corresponding optical periods are on the order of 35–42 fs.

```
T = 1 / f
  = 1 / (2.4e13 Hz) = 4.2e-14 s = approx. 42 fs   (near TO)
  = 1 / (2.9e13 Hz) = 3.4e-14 s = approx. 34 fs   (near LO)
```

In a polariton-to-polariton architecture, mid-IR light is confined to coupling energy into and out of the structure (via antennas, gratings, prisms, or near-field probes) and to readout; it must not perform the Boolean operation itself. Because surface phonon-polaritons are hybrid light–lattice excitations, the distinction between an optical field and a polariton state must be made explicit in any claimed demonstration.

**Distinct roles:**

```
1. Polariton input preparation: launches calibrated input
   polariton states into the SiC structure.
2. Polariton logic region: performs the polariton-polariton
   Boolean interaction.
3. Polariton readout: measures output amplitude, phase, or
   near-field intensity.
```

A polariton-to-polariton gate requires:

```
polariton input A --+
                    +--> polariton-polariton logic --> output polariton
polariton input B --+
```

It must not be:

```
mid-IR source A --+
                  +--> optical excitation --> detector readout
mid-IR source B --+
```

unless the optical stage is explicitly only state preparation and the Boolean decision occurs between polariton states.

Patterned SiC resonators, pillars, gratings or laser-modified regions may affect confinement, scattering, resonance frequency, and loss, but no specific SiC pattern is established as a working SHA-256 gate. Begin with unpatterned controls, simple resonator test structures, and checks for heating, damage, and drift.

## 7. Important Physical Logic Limitations

Linear polariton superposition alone does not implement AND, NAND, or NOT. A convincing polariton-to-polariton gate needs a reproducible nonlinear interaction or justified interference/readout mechanism, with every truth-table row measured using polariton inputs and a polariton-domain output.

If mid-IR pulses encode the inputs and a detector immediately performs the Boolean decision, the system is optical-to-polariton or polariton-to-optical, not a complete polariton-to-polariton

logic gate. External electronics or software must not perform the Boolean decision.

A convincing demonstration should quantify output error rate, repeatability, latency, signal separation, drift, thermal effects, sample variation, and cascading behaviour between gates. Propagation loss and finite polariton lifetime (typically picosecond-scale in SiC) set hard limits on circuit depth without regeneration.

## 8. Experimental Checklist

- [ ] Define polariton-domain encoding of 0 and 1.
- [ ] Define polariton output measurement and classification threshold.
- [ ] Prepare two independent input polariton states.
- [ ] Measure every polariton-input truth-table row.
- [ ] Demonstrate polariton-input NOT: 0 -> 1 and 1 -> 0.
- [ ] Include no-input, single-input, and unpatterned-sample controls.
- [ ] Quantify repeatability and polariton output error rates.
- [ ] Check heating, damage, polariton loss/decoherence, and signal drift.
- [ ] Demonstrate polariton-input XOR.
- [ ] Demonstrate polariton-input full-adder SUM and COUT.
- [ ] Verify polariton carry propagation through multiple bit positions.
- [ ] Verify 32-bit modular addition using polariton-domain words.
- [ ] Verify rotations and shifts as polariton bit routing.
- [ ] Compare each stage with a software reference.
- [ ] Test standard SHA-256 known-answer vectors.
- [ ] Keep measured evidence separate from conceptual designs.

## 9. Summary

The core SHA-256 building blocks are NOT, AND, XOR, OR or equivalent derived logic, 32-bit modular addition with carry propagation, and fixed rotations/shifts. Choice and Majority are Boolean combinations; Choice explicitly requires NOT. Sigma functions combine rotations or shifts with XOR. For a proposed SiC phonon-polariton implementation, every logic operation should take polariton states as inputs and produce a polariton state as output; mid-IR light should be limited to coupling, preparation and measurement.

## 10. Polariton-to-Polariton Physical Gate Architectures

### 10.1 Polariton Encoding and Readout

A proposed interface uses a patterned 4H- or 6H-SiC resonator array (or SiC film on a suitable substrate) excited in the mid-IR Reststrahlen band. Logical 0 is a null or sub-threshold polariton amplitude; logical 1 is a calibrated polariton amplitude or phase state at a chosen resonance. Inputs and outputs are polariton states. Readout may use far-field spectroscopy or scattering-type near-field microscopy, but it must only read the result; it must not create the Boolean output.

```
polariton A --+
              +--> polariton-polariton logic node --> polariton Y --> readout
```

```
polariton B --+

Encoding:
  0 = null / sub-threshold polariton amplitude
  1 = calibrated polariton amplitude or phase state
```

## 10.2 Polariton-to-Polariton NOT Gate

A polariton-input NOT gate must convert a null or low-amplitude input into a calibrated high-amplitude output, and convert a high-amplitude input into a null or low-amplitude output. Candidate mechanisms include a biased interferometer with a constant reference polariton, saturable or nonlinear resonance shifting, phase-sensitive interference, or nonlinear mode conversion whose output amplitude is the complement of the input state. A NOT gate requires gain or a stored bias: a passive linear structure cannot turn zero input into non-zero output.

**Polariton NOT truth table:**

| Input A | Output NOT A |
|---|---|
| 0 | 1 |
| 1 | 0 |

The inversion must occur in the polariton interaction region. A probe may measure the output, but must not supply the missing logical state.

## 10.3 Polariton-to-Polariton AND Gate

A polariton-to-polariton AND gate requires two independently prepared polariton inputs, A and B, to interact in a shared SiC region. The proposed hypothesis is that a nonlinear interaction produces a distinguishable output state only when both inputs are present above threshold.

```
polariton A --+
              +--> nonlinear polariton interaction --> polariton Y
polariton B --+
```

**Required result:**

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

Candidate mechanisms include thresholded nonlinear resonance shifting, second-order three-wave (sum- or difference-frequency) mixing, parametric mixing, or a nonlinear scattering process that generates a detectable output mode only for the joint high-input case. These are hypotheses, not established SiC AND-gate designs.

Common SiC polytypes such as 4H and 6H are non-centrosymmetric and have a non-zero second-order susceptibility, and the lattice has anharmonic phonon–phonon coupling. However, nonlinear response in SiC polariton resonators is generally weak at low intensity, and anharmonic decay is typically a loss process; neither automatically implements the AND truth table. The achievable nonlinear contrast at energies compatible with a dense circuit must be established by measurement or validated simulation.

A valid demonstration must show that the output is a polariton state generated by the two polariton inputs, not merely a reflectance or absorption change caused by two optical beams. The readout must measure the output after the polariton interaction.

## 10.4 Polariton-to-Polariton XOR Gate

A polariton-to-polariton XOR gate requires two polariton inputs to interfere or undergo phase-sensitive conversion such that one input produces a high output, while two inputs with a designed relative phase produce cancellation or a distinguishable low output.

```
polariton A --+
              +--> phase-sensitive polariton interferometer --> polariton Y
polariton B --+
```

**Required result:**

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

Candidate implementations include a Mach-Zehnder-style polariton interferometer, a coupled-resonator or directional-coupler structure with phase-dependent output ports, or nonlinear mixing with a bias field. The relative phase must be controlled between polariton packets in the device, not merely between external optical beams.

## 10.5 Polariton-to-Polariton Full Adder

A full adder must accept three polariton-domain inputs: A, B, and carry-in. It must produce two polariton-domain outputs: SUM and carry-out. The SUM stage can use cascaded polariton XOR interactions; the carry stage can use polariton AND interactions followed by a polariton OR-equivalent threshold or mode-combining interaction.

```
SUM  = A XOR B XOR Cin
COUT = (A AND B) OR (Cin AND (A XOR B))

polariton A   --+
polariton B   --+--> XOR stage --> XOR stage --> SUM
polariton Cin --+

polariton A   --+
polariton B   --+--> AND --> OR-equivalent --> COUT
polariton Cin --+--> XOR --> AND ------------/
```

All three inputs and both outputs must remain polariton states throughout the computation. Mid-IR light may couple input polaritons in and read output polaritons out, but the adder logic itself must be polariton-polariton.

## 10.6 Polariton Routing and Cascading

A complete SHA-256 processor would require polariton waveguides, delay lines, splitters, phase shifters, mode converters, and regeneration or amplification stages. Rotations and shifts may be implemented as polariton routing between bit positions. Carry propagation requires a chain of polariton full adders with controlled timing and low loss.

Key open problems include propagation loss, finite polariton lifetime, thermal crosstalk, fan-out, mode and impedance matching, in/out-coupling efficiency, fabrication tolerance, timing jitter, and whether a weak nonlinear interaction can cascade without optical or electronic regeneration. These issues must be measured before claiming a scalable polaritonic SHA-256 circuit.

# 11. Resistive Input Handler for the 5 × 5 µm Engraved SiC Field

## 11.1 Assumptions and scale check

This section assumes the 5 × 5 µm region is the e-beam-patterned *active field* on a larger SiC die or carrier. A 5 µm piece cannot carry bond pads, so all electrical contacts, fan-out traces and optical coupling structures sit outside the field and connect to it.

Scale matters. Free-space mid-IR wavelengths in the Reststrahlen band (about 10–12 µm) are larger than the field itself, but surface phonon-polaritons have a much shorter wavelength than the light that excites them, which is what makes micron-scale resonators conceivable. Even so, a 5 × 5 µm field can plausibly hold only a few resonator elements up to a small gate network. A single SHA-256 round needs on the order of thousands of two-input gate equivalents, before counting the message schedule. The field described here is therefore a test structure for Stages 1–4 (single gates, XOR, a full adder), not a SHA-256 processor.

## 11.2 Role of the input handler

The input handler loads data into the device. It formats and delivers bits; it does not compute the hash. Message padding and splitting into 512-bit blocks are data formatting done on the host, and the initial hash values and round constants are fixed data. All Boolean evaluation under test must still happen in the polariton domain, and the electronics must not decide any truth-table output.

```
host software (padding, 512-bit block -> 16 x 32-bit words)
        |
        v
serial loader / shift register + latch (carrier board or CMOS)
        |
        v
per-channel current driver (DAC or calibrated pulse source)
        |
        v
fan-out metal traces  --->  resistive elements at the engraved field
                                      |
                                      v
                        thermally set input state per channel
                                      |
                    (IR stimulator launches the polariton: Section 13)
```

## 11.3 Resistive element and encoding

A resistive element (Joule heater) produces incoherent thermal phonons, not coherent polaritons. It cannot launch the polariton signal. Its useful role here is as a **state-setting input**: a calibrated local temperature rise shifts a resonator's phonon-dependent response relative to the fixed stimulator frequency, or acts as a thermal phase shifter in a waveguide arm.

```
Encoding (per input channel):
  0 = baseline bias current (resonator at reference temperature)
```

```
1 = calibrated current pulse giving target temperature rise dT
    (resonance detuned toward or away from the stimulator line)
```

Candidate heater types are a patterned metal microheater placed beside the resonator, or a doped-SiC resistive channel. Metal near a polariton mode adds absorption and perturbs the mode, so spacing and layout must be simulated. Doped SiC adds free-carrier damping to the optical response. Either choice must be checked against the loss budget.

| Parameter | Consideration (values to be verified) |
|---|---|
| Tuning sensitivity | Raman studies report SiC optical-phonon lines shifting by roughly hundredths of a $cm^{-1}$ per kelvin. Polariton resonator linewidths are of order 10 $cm^{-1}$, so a full-linewidth shift may need a temperature rise of hundreds of kelvin. |
| Speed | Thermal diffusion time scales as $L^2/\alpha$. With L = 5 µm and $\alpha \approx 1.7$ $cm^2/s$ for SiC this is of order 100 ns, so bit-setting is MHz-class, many orders slower than the 35–42 fs polariton period. |
| Crosstalk | SiC conducts heat very well, so heat from one channel spreads quickly to neighbours at micron pitch. Channel spacing and thermal isolation need modelling. |
| Fan-out | Pads are tens of µm across; the field is 5 µm. Routing density limits how many input channels can reach the field. |
| Readback | Heater resistance (via its temperature coefficient) can be sensed with a four-point layout to confirm the set temperature before the stimulator fires. |
| Damage / drift | Metal traces limit current density (electromigration). Check heating, resonance drift and sample-to-sample variation. |

### 11.4 Timing sequence

1. Load bits into the shift register and latch.
2. Apply bias/pulse currents to the resistive elements.
3. Wait for thermal settling; verify by resistance readback.
4. Gate the IR stimulator on (synchronised, low duty cycle).
5. Open the readout window; record amplitude or phase.
6. Classify against the preset threshold; log the result.
7. Return heaters to baseline; wait for cool-down; next vector.

### 11.5 Limits to state in any report

Thermally set inputs are electrical-to-thermal-to-polariton, not polariton inputs. Any claim of a polariton-to-polariton gate must say which parts are state preparation and show that the Boolean interaction itself occurs between polariton states. A heater-tuned resonator on its own is a thermally switched filter, and demonstrating logic needs the full truth table with the controls listed in Section 8.

## 12. Laser Spectroscopy and Readout

### 12.1 Principle

Laser spectroscopy reads the device state. It must measure outputs and must not supply the logical result. Spectroscopy also checks that the input handler set the intended state, since the heater temperature and the polariton response are both measurable.

### 12.2 Candidate methods

| Method | What it measures | Notes |
|---|---|---|
| Raman micro-spectroscopy (visible laser, e.g. 532 nm) | Lattice phonon line position and width, giving local temperature and strain. | Spot near 1 µm can resolve structure inside the 5 × 5 µm field. Visible light is below the 4H-SiC bandgap (about 3.2 eV), so absorption is low. It reads lattice phonons, not the polariton output directly. Selection rules depend on polytype. |
| s-SNOM with tunable QCL or $CO_2$ laser (about 10–11 µm) | Near-field amplitude and phase of the surface polariton, at tip-limited resolution (tens of nm). | Best match for polariton mapping at this scale. $CO_2$ laser lines fall inside the SiC Reststrahlen band. The tip perturbs the mode and scanning is slow, so this is characterisation rather than in-line readout. |
| Far-field FTIR or QCL reflectance microscopy | Resonance spectrum (frequency, linewidth, depth). | Diffraction-limited spot is comparable to or larger than the field, so it averages over the whole field. Normalise against an adjacent reference region. |
| Mid-IR pump–probe (optional) | Polariton dynamics and lifetime. | Needs ultrafast mid-IR sources; useful for loss and timing budgets. |

### 12.3 Readout chain and thresholding

```
tunable laser (stabilised) --> modulator/chopper --> polarisation optics
  --> reflective objective --> sample (Locale R)
  --> cooled MCT detector --> lock-in / digitiser
  --> threshold classifier --> record result

Reference arm: unpatterned SiC region read simultaneously.
```

For logic readout, fix the laser near a resonance and measure the amplitude or phase change, rather than sweeping the spectrum each time. Sweeps are used to locate and calibrate the resonance. Choose thresholds with margin from the measured noise distributions of the 0 and 1 states.

### 12.4 Calibration and controls

Include unpatterned-SiC, heater-only (no stimulator), stimulator-only (no heater), and power-sweep controls. Wait for thermal equilibrium before each reading, repeat vectors to estimate the error rate, and separate heating-induced reflectance shifts from a genuine polariton output change.

## 13. Infrared Laser Stimulator and the Second Coupling Locale

### 13.1 Why a separate locale

The stimulator is coupled in at a different location (Locale S) from the readout (Locale R). A focused mid-IR spot is at best a few micrometres across (roughly λ/2 at high numerical aperture, more in practice), so it would overfill a 5 × 5 µm field and illuminate the heaters and traces directly. A larger coupler next to the field launches a surface polariton that is guided to the logic region. Separate locales also keep stimulator light out of the detector.

```
[Locale S: stimulator coupler]
      |  SPhP taper / guide
      v
[5 x 5 um engraved logic field + resistive inputs]
```

```
      |  output guide
      v
[Locale R: out-coupler / s-SNOM / far-field readout]

[Reference region]   [heater pads and fan-out traces, off-field]
```

### 13.2 Coupling options at Locale S

| Option | How it works | Trade-off |
|---|---|---|
| Grating coupler | Periodic grooves supply the missing momentum: $k_{SPhP} = k_0 \sin\theta + 2\pi m/\Lambda$. | Simple and planar. Period $\Lambda$ must come from simulated dispersion; efficiency is typically modest. |
| Antenna / resonator-fed launch | A resonant structure scatters incident light into the guided polariton. | Compact and narrow-band; sensitive to fabrication tolerance. |
| Prism (Otto geometry) | Evanescent coupling across a small gap. | Efficient but awkward to align with a micro-scale chip. |
| s-SNOM tip launch | The tip supplies momentum locally. | Good for experiments; not a fixed on-chip coupler. |

### 13.3 Stimulator requirements

Surface phonon-polaritons are transverse-magnetic, so the stimulator must be p-polarised relative to the coupler. A tunable QCL (about 8–12 µm) or a $CO_2$ laser line near 10.6 µm are candidate sources. The linewidth should be well below the resonance width (target below about 1 $cm^{-1}$ against roughly 10 $cm^{-1}$). Set the frequency slightly off the cold-state resonance so the heater-set state moves the device toward or away from it. Keep average power low and use gated, modulated operation synchronised with the input handler, because stimulator heating can mimic an input.

### 13.4 Isolation, alignment and stability

Separate the stimulator from the readout by position, polarisation and time gating. If Raman is used for readout, its visible wavelength is spectrally distinct from the mid-IR stimulator. Engrave alignment marks next to each locale. Align the stimulator spot to the coupler to within about a micrometre, and track drift. Confirm the response is in the linear regime with a power sweep, and measure coupling efficiency and propagation loss between Locale S and the field rather than assuming them.

## 14. Added Checklist for Sections 11–13

- [ ] Confirm active-field size, layout and how many gate elements fit.
- [ ] Calibrate heater current versus temperature and resonance shift.
- [ ] Measure thermal settling time and neighbour-channel crosstalk.
- [ ] Verify the host loader delivers the intended bit pattern (no Boolean decisions in electronics).
- [ ] Characterise coupling efficiency and loss, Locale S to field to Locale R.
- [ ] Map the polariton mode (s-SNOM) and confirm the resonance spectrum.
- [ ] Run heater-only, stimulator-only and unpatterned controls.
- [ ] Check linearity and heating from the stimulator by power sweep.
- [ ] Measure the full truth table with thresholds fixed in advance.

[ ] Report which stages are state preparation and which are polariton-domain logic.

***Disclaimer:*** *These notes are educational and speculative. They do not demonstrate, fabricate, or validate a physical silicon carbide phonon-polariton SHA-256 processor.*